



Islamic University of Technology
Department of Computer Science and Engineering

Lab 3: Graph Traversal

CSE 4404: Algorithms Lab(SWE)
Summer 2024-25

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Task A. Patient Zero

Time Limit: 1 second | Memory Limit: 256 MB

A city is modelled as an $n \times m$ grid. Each cell is either . (an open street), # (a wall that cannot be entered), or Z (the single starting zombie — Patient Zero). Every minute, the zombie spreads to all adjacent open streets (up, down, left, right). Walls block the spread.

For each cell, determine the minimum number of minutes until a zombie reaches it. If a cell is unreachable, output -1 . The starting zombie cell should be output as 0. Wall cells (#) should be output as -1 .

Input Format

The first line contains a single integer t ($1 \leq t \leq 100$) — the number of test cases.

For each test case:

- The first line contains two integers n and m ($1 \leq n, m \leq 1000$) — the number of rows and columns.
- The next n lines each contain a string of length m consisting only of ., #, and Z. There is **exactly one** Z in each test case.

It is guaranteed that the sum of $n \times m$ over all test cases does not exceed 10^6 .

Output Format

For each test case, print n lines each containing m space-separated integers — the minimum time for the zombie to reach each cell, or -1 if unreachable or a wall.

Examples

Sample Input	Sample Output
2	0 1 2
3 3	1 2 3
Z..	2 3 4
...	5 4 3 2
...	4 3 2 1
3 4	-1 2 1 0
....	
....	
#..Z	

Notes

In the first test case, the zombie starts at (0,0) and spreads outward. With no walls, every cell is reachable and the time equals the Manhattan distance from the starting cell.

In the second test case, the zombie starts at (2,3) (bottom-right corner). The wall at (2,0) makes that cell unreachable. All other open cells are reachable, with distances increasing as you move away from the starting cell.

Task B. Broadcast Storm

Time Limit: 1 second | Memory Limit: 256 MB

A university has n computers connected by m bidirectional network cables. Computers that are directly or indirectly connected form a **subnet**. A subnet that contains a cycle causes a **broadcast storm** — packets loop through the cycle forever, crashing the entire subnet.

Given the network layout, count how many subnets will experience a broadcast storm.

Input Format

The first line contains a single integer t ($1 \leq t \leq 10^4$) — the number of test cases.

For each test case:

- The first line contains two integers n and m ($1 \leq n \leq 2 \times 10^5$, $0 \leq m \leq 5 \times 10^5$) — the number of computers and cables.
- The next m lines each contain two integers u and v ($1 \leq u, v \leq n$, $u \neq v$) — a bidirectional cable between computers u and v .

The graph is simple (no self-loops or duplicate cables). The sum of n over all test cases does not exceed 2×10^5 , and the sum of m does not exceed 5×10^5 .

Output Format

For each test case, print a single integer — the number of subnets that contain a cycle.

Examples

Sample Input	Sample Output
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3	2
8 9	0
1 2	2
2 3	
3 1	
1 4	
4 5	
5 1	
6 7	
7 8	
8 6	
4 3	
1 2	
2 3	
3 4	
6 6	
1 2	
2 3	
3 1	
4 5	
5 6	
6 4	

Notes

In the first test case there are two subnets. The first subnet $\{1, 2, 3, 4, 5\}$ contains a cycle, and so does the second subnet $\{6, 7, 8\}$. Both subnets are cyclic. Answer: 2.

In the second test case, the four computers form a simple path 1-2-3-4 with no cycle. Answer: 0.

In the third test case, computers $\{1, 2, 3\}$ form one subnet (triangle) and computers $\{4, 5, 6\}$ form another subnet (triangle). Both subnets have a cycle. Answer: 2.

Marks Distribution

Task	Marks
Task A	50%
Task B	50%

Practice Problems

Problem	Links
Counting Rooms	CSES: https://cses.fi/problemset/task/1192
Labyrinth	CSES: https://cses.fi/problemset/task/1193
Building Roads	CSES: https://cses.fi/problemset/task/1666
Learning Languages	Codeforces: https://codeforces.com/contest/277/problem/A Vjudge: https://vjudge.net/problem/CodeForces-277A
Kefa and Park	Codeforces: https://codeforces.com/contest/580/problem/C Vjudge: https://vjudge.net/problem/CodeForces-580C